

E-commerce Sales Dashboard: Building an Interactive View of Online Store Performance

This report walks through the creation of an interactive e-commerce sales dashboard built from an online store's order data. The work centered on a dataset of online orders containing product, pricing, discount, payment, and purchase date details, which was used to generate a full year-and-a-half of daily sales activity and power a suite of connected, filterable visualizations. Two versions were produced: a dark analyst-style dashboard and a cleaner executive edition, both offering live filters so users can slice the numbers by their own questions.

The Foundation: A Real Order Dataset

The starting point was an uploaded order file covering everyday online retail activity. Each row captured a customer, a product, its category, the original price, the discount applied, the final price paid, the payment method, and the date of purchase.

- The source dataset held **3,660 orders** across familiar shopping categories such as Sports, Clothing, Toys, Beauty, and Home & Kitchen.
- Prices ranged widely, from budget items around Rs. 36 to higher-ticket purchases above Rs. 300, with discounts commonly between 15% and 25%.
- Payments came through several methods, including Net Banking, Credit Card, Debit Card, and UPI.

A sample of the original records:

User_ID	Category	Price (Rs.)	Discount (%)	Final_Price (Rs.)	Payment Method
337c166f	Sports	36.53	15	31.05	Net Banking
d38a19bf	Clothing	232.79	20	186.23	Net Banking
d7f5f0b0	Sports	317.02	25	237.76	Credit Card
395d4994	Toys	173.19	25	129.89	UPI
a83c145c	Beauty	244.80	20	195.84	Net Banking

Expanding the Picture: Daily Activity Over Time

To give the dashboard meaningful trends, the order data was extended into a daily series running from **January 1, 2024 through June 20, 2025**. This expansion added the dimensions a sales team typically wants to track but that were missing from the raw file: customer acquisition channels, geographic regions, and a shopping funnel from product view to purchase.

- Acquisition channels included Organic Search, Paid Social, Email, Direct, Affiliate, and Referral, with Organic Search and Direct as the largest sources of traffic.
- Sales were distributed across five regions — North, South, East, West, and Central.
- A built-in seasonal rhythm and a gentle upward growth trend were layered into daily order counts, so revenue rises modestly over the period rather than staying flat.

A sample of the expanded daily data:

Date	Category	Price	Discount (%)	Final Price	Payment Method
2024-01-01	Toys	39.18	6.70	36.55	Debit Card
2024-01-01	Home & Kitchen	253.94	46.58	135.66	UPI
2024-01-01	Sports	257.06	7.84	236.90	Net Banking
2024-01-01	Sports	625.38	11.19	555.39	Debit Card
2024-01-01	Beauty	129.39	0.00	129.39	Debit Card

What the Dashboard Shows

The dashboard brings all of these threads together into a single screen, with each chart answering a specific business question.

- **Revenue and order volume trends** sit at the top, plotting monthly revenue against order counts so peaks and slow periods are easy to spot.
- **Average order value over time** tracks whether the typical basket is growing or shrinking month to month.
- **Top products by revenue** ranks the ten best-selling items, making it clear where the money concentrates.

- **Customer acquisition channels** appear as a share-of-orders breakdown, highlighting which channels bring in the most activity.
- **A cart abandonment funnel** follows shoppers through four stages — Viewed Product, Added to Cart, Checkout Started, and Purchased — exposing where buyers drop off.
- **A geographic sales heatmap** cross-references regions against categories to reveal where specific product types sell best.
- **A real-time KPI scoreboard** anchors the view with four headline numbers: total Revenue, total Orders, Average Order Value, and Conversion rate.

Every element updates together. As filters change, the KPI cards and all six charts recalculate instantly to match the selection.

Two Versions for Two Audiences

The first dashboard used a dark navy theme suited to hands-on analysis, with dropdown filters for category, region, and channel plus a product search box.

The second version was redesigned for a boardroom audience, using a clean white-and-steel palette, a KPI-first layout, and a "Live filters" label in the header. It added the requested **date range** and **payment method** slicers alongside the product search, so leaders can zoom into a specific quarter or compare how different payment types perform.

Both versions were ultimately saved as standalone HTML files so they open reliably:

- Analyst version: `ecommerce_dashboard_v1.html`
- Executive version: `executive_ecommerce_dashboard.html`

Summary of Key Insights

The project transformed a simple 3,660-row order file into two fully interactive sales dashboards covering nearly eighteen months of activity. The raw data provided the commercial backbone — products, prices, discounts, payment methods — while the expanded daily series added the channel, regional, and funnel dimensions needed for a complete operating view. The resulting dashboards let any user move from a high-level KPI scoreboard down to specific products, regions, channels, date ranges, and payment methods, with revenue trends, average order value, top sellers, acquisition mix, the purchase funnel, and a geographic heatmap all responding to the same filters in real time. Offered in both an analyst-friendly dark theme and a polished executive layout, the dashboard serves both deep exploration and quick leadership review from the same underlying data.